KAPPA FOUNDATION

Initial Decision Record | Case decision: 31 December 2022

Investment decision

Kappa received $12 million in cash. We first allocate default-free case bonds to the $2.4 million scholarship payment due 31 December 2024 and the $3.6 million learning-centre payment due 31 December 2027. The remaining capital is invested for long-term growth. The goal is to retain the purchasing power of $8 million (December 2022 dollars) at end-2030, not to guarantee this outcome.

We recommend Sample Mean-Variance (MV) for the equity sleeve, with a 15% cap per common stock and a 30% cap per direct-stock sector. A constrained Minimum-Variance portfolio under the same limits is the rejected but viable alternative. No borrowing, short sales or derivatives are permitted.

Initial allocation and funding

Asset / use	Initial allocation
2,400 Beta zero-coupon bonds, principal due 2024	$2,214,672.00
3,600 Delta 3.5% coupon bonds, principal due 2027	$3,549,852.00
Growth equity securities (net purchase amount)	$6,229,246.75
One-time equity entry costs, 10 basis points	$6,229.25
Operating cash buffer	$0.00
Total	$12,000,000.00

Beta principal pays the 2024 liability; Delta principal pays the 2027 liability. Delta pays $126,000 annually during 2023-2027; the $630,000 total is retained as cash earning 0% through 2030 in the base case. These coupons are not counted twice as investment income. The initial $0 cash buffer is a deliberate trade-off, not a promise of ready liquidity.

Equity universe and chosen weights

Candidates: AAPL, MSFT (technology); JPM, BAC (financials); JNJ, PFE (health care); PG, KO (consumer staples); XOM (energy); CAT (industrials). ETFs: VTV (US large-cap value) and IWM (US small-cap). The tradable benchmark is VTI. The optimizer may give eligible assets zero weight.

Ticker	Share of growth sleeve	Purchase amount
AAPL	15.00%	$934,387.01
MSFT	15.00%	$934,387.01
JPM	4.24%	$264,259.39
BAC	0.00%	$0.00
JNJ	9.10%	$566,729.50
PFE	14.17%	$882,725.07
PG	15.00%	$934,387.01
KO	12.59%	$784,421.14
XOM	0.00%	$0.00
CAT	14.90%	$927,950.62
VTV	0.00%	$0.00
IWM	0.00%	$0.00

Dollar purchases sum to $6,229,246.75; full-precision weights, including zero allocations, and security eligibility records are in the supporting data. The eight positive stock positions span five sectors.

Portfolio estimation and evidence

We use 96 monthly Yahoo Finance adjusted-price returns (January 2015-December 2022), retrieved retrospectively using yfinance. Forecast annual arithmetic mean = 12 × [0.25 × asset historical mean + 0.75 × VTI historical mean]. Annual sample covariance = 12 × monthly sample covariance. The 12-asset universe and all constraints are held constant when we compare sample and VTI single-index risk models.

The simplified foundation utility contribution from equities is f × expected growth return minus (A/2) × f^2 × growth variance, where f = $6,229,246.75 / $12 million and foundation A = 3. A = 6 is a sensitivity check. The calculation does not include complete bond-equity covariance or dynamic withdrawals.

Growth strategy	Expected return	Volatility	Equity utility, A=3
Sample MV (selected)	12.05%	14.98%	0.053469
Single-index MV	11.91%	14.46%	0.053348
Minimum Variance (rejected)	10.86%	13.15%	0.049374

Sample MV has the highest estimated utility for our stated inputs, but not a statistically established future advantage. The selected growth portfolio changes when assumptions change: for the two fixed Sample MV and Minimum-Variance portfolios, the preference reverses near historical-mean weight 0.0842 (versus our 0.25 assumption), or A near 8.91. Reoptimization leads to different thresholds.

Downside risk and purchasing power

The trustees' 15% loss concern is interpreted as a mandatory review threshold, not a maximum loss guarantee. A trailing 12-month, withdrawal-adjusted modeled whole-fund loss of 12% prompts an early warning; 15% requires investment-committee review within ten business days. Mandatory programme payments are external withdrawals, not investment losses. Because the bonds are fictional, post-2022 bond market values require explicitly modeled rates; they cannot be presented as observed quotes. Growth-only returns and funding coverage are monitored separately.

Our adverse tests show about a 17.27% immediate decline in initial fund value under an equity market fall of 30% combined with a 100-basis-point rise in Beta/Delta yields, and about 16.31% under a selected sector selloff with the same rate shock. These are illustrative shocks, not one-year return forecasts. Minimum Variance has lower estimated volatility. At 2.3% inflation, the December 2030 goal is about $9.596 million nominal. With $630,000 retained coupon cash earning 0%, the growth sleeve needs roughly 4.66% geometric CAGR. No return is guaranteed.

Implementation and original evaluation rule

Purchase fractional equity positions at the 3 January 2023 closing-price proxy, after the 10 bp entry cost. Do not make discretionary trades or rebalance to target weights. Model automatic reinvestment of each security's cash dividends in that security using adjusted-close total-return proxies: portfolio weights drift and actual share counts may rise. Process mandatory corporate actions consistently; do not elect voluntary tender or exchange offers. Check vendor adjustments when evaluating outcomes, and do not count dividends twice. Use the same entry date and fee for VTI; January returns begin at the 3 January close, not the previous December close.

Future evaluation reports growth-only compound return, volatility, maximum drawdown and active return against the frozen VTI benchmark. Bond funding is analyzed through contractual cash flows and scenarios. The original decision uses no 2023-2025 returns. These policy and evaluation rules will be included in the 22 October submission, before the revised mandate. Any later change must be documented separately with its date and reason.